Heart Disease Analysis

Phase 3: Project Design Phase

Solution Architecture

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Architecture Overview

The system follows a simple layered architecture: a data storage layer (MySQL), an analysis layer (SQL), a visualization layer (Tableau), a publishing layer (Tableau Public), and a presentation layer (Flask web app).

System Architecture Flow

[Data Source: Heart_new2.csv (Kaggle dataset)]

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[Storage: MySQL — heart_disease_db.HD]

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[Analysis: SQL Queries (aggregation, grouping, filtering)]

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[Visualization: Tableau Desktop (Extract-based data source)]

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[Presentation Assets: Dashboard + 6-part Story]

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[Publishing: Tableau Public (hosted, shareable view)]

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[Web Layer: Flask App (iframe embed + custom UI)]

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[Access: End User Browser]

Component Responsibilities

Component	Responsibility
MySQL Database	Persistent structured storage of the patient dataset; supports SQL-based analysis.
SQL Analysis Scripts	Compute prevalence, demographic, and risk-factor metrics used to drive visuals.
Tableau Desktop	Build calculated fields, charts, dashboard, and story; convert live connection to extract for portability.
Tableau Public	Hosts the published, publicly accessible version of the dashboard/story.
Flask Application	Serves a lightweight web page embedding the Tableau view via iframe.

Component	Responsibility
GitHub Repository	Stores dataset, SQL scripts, Tableau workbook, Flask app code, and documentation.

Deployment View

Development: MySQL and Tableau Desktop run locally during data preparation and dashboard building.

Publishing: The finished Tableau workbook (using an extract, not a live DB connection) is published to Tableau Public.

Hosting: The Flask app, containing only the embed code and minimal UI, is deployed to a free hosting service such as Render or PythonAnywhere.

Access: End users reach the dashboard either directly via the Tableau Public link, or through the Flask app's embedded view.

Design Considerations

Because Tableau Public does not support live database connections, the architecture deliberately converts the MySQL-backed data source into a Tableau Extract before publishing — decoupling the published dashboard from the live database and making it portable to Tableau Public's free hosting.